

RIAZ ASLAM

Software Engineer | AI/ML & Full-Stack Development

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PROFESSIONAL SUMMARY

Third-year Software Engineering undergraduate (CGPA 3.63) with two internships, 20+ public repositories, and a Top 10 finish in the global Build with Kiro Hackathon. Builds end-to-end systems across React, Next.js, FastAPI, and PostgreSQL, with applied depth in machine learning (Scikit-Learn, XGBoost), LLM integration (Google Gemini, Amazon Bedrock), and low-level storage engineering (write-ahead logging, B-Tree indexing). Seeking a Software Engineering or AI/ML internship for 2026/2027.

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

Frontend: React.js, Next.js, Angular, Tailwind CSS, Responsive UI, REST API Integration

Backend & Databases: FastAPI, Node.js, Express.js, PostgreSQL, SQL Server, MySQL, DuckDB, RESTful APIs, RBAC Auth, 3NF Normalization

AI & Machine Learning: Scikit-Learn, XGBoost, Pandas, NumPy, Google Gemini API, Amazon Bedrock, Prompt Engineering

Systems & CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Write-Ahead Logging (WAL), B-Tree Indexing, Concurrency, SDLC

Tools & Platforms: Git, GitHub, Docker, Vercel, Postman, VS Code, IntelliJ IDEA, Qt Framework, Streamlit

PROFESSIONAL EXPERIENCE

AI/ML Intern | FlyRank | May 2026 - Aug 2026

- Developed and evaluated machine learning models for ranking and recommendation workflows using Python, Scikit-Learn, and XGBoost.
- Built data preprocessing and feature-engineering pipelines with Pandas and NumPy, improving dataset quality and training consistency.
- Served trained models through REST endpoints and documented experiment workflows so results were reproducible across the team.

Volunteer Management Intern | Alkhidmat Foundation Pakistan | Mar 2026 - Aug 2026

- Coordinated volunteer operations and logistics for large-scale youth mobilization drives across multiple campuses.
- Maintained volunteer database records and reporting, reducing manual tracking effort and improving data accuracy for event planning.

Campus Ambassador | Manafa (Software House) - COMSATS University, Lahore | Oct 2025 - Present

- Official on-campus representative, leading student outreach, recruitment drives, and technical engagement across engineering cohorts.

Social Co-Lead | Google Developer Groups on Campus (GDGoC), COMSATS Lahore | Sep 2025 - Present

- Co-lead developer outreach, technical workshops, and hackathons with the core organizing team, driving higher student participation in hands-on coding sessions.

TECHNICAL PROJECTS

Acadexa AI - AI Academic Management Platform | React, Next.js, FastAPI, PostgreSQL, Google Gemini API

- Built a multi-tenant platform automating curriculum parsing, student attendance, quiz generation, and grading analytics.
- Integrated the Google Gemini API for sub-second AI responses and designed a normalized PostgreSQL schema enforcing tenant data isolation.

CacheForge - Zero-Dependency Embedded Storage Engine | Python, WAL, B-Tree Indexing, ACID

- Engineered an ACID-compliant embedded key-value engine with write-ahead logging, B-Tree indexing, TTL expiration, and crash recovery, using no external databases or third-party libraries.
- Implemented journal replay to guarantee durable state restoration after simulated process failures.

RankPilot AI - Search Intelligence & Recommendation Platform | Python, XGBoost, DuckDB, FastAPI, Streamlit

- Trained gradient-boosted ranking models and served them through a FastAPI backend with a Streamlit analytics dashboard.
- Leveraged DuckDB columnar OLAP storage to query and rank large datasets at sub-millisecond latency.

Personal AI Study Coach (Kiro) - Top 10, Build with Kiro Hackathon | TypeScript, React, Next.js, Node.js, Tailwind CSS

- Developed an adaptive study companion generating personalized task roadmaps and tracking focus sessions against learning progress.
- Shipped a production deployment on Vercel; selected in the global Top 10 from the competition field.

Graph-Based Social Network Engine | C++, Qt Framework, SFML, Data Structures

- Implemented custom hash tables, priority queues, and BFS/DFS graph traversal with zero-library file persistence, prioritizing memory efficiency and predictable lookup performance.

Smart Inventory & Sales Management Platform | Angular, Node.js, Express, SQL Server

- Delivered a full-stack inventory system with RBAC authentication, REST APIs, real-time stock alerts, and sales dashboards.

EDUCATION

BS Software Engineering - CGPA 3.63 / 4.00 | COMSATS University Islamabad, Lahore Campus | 2024 - 2028 (Expected)

- Relevant Coursework: Data Structures & Algorithms, OOP, Database Systems, Operating Systems, Computer Networks, Software Design & Architecture, Machine Learning.

FSc Pre-Engineering - 1081 / 1100 (A+) | KIPS College, Lahore | 2021 - 2023

ACHIEVEMENTS & CERTIFICATIONS

- Top 10 Finalist** - Build with Kiro Hackathon, for an AI-powered adaptive study coaching platform.
- Participant** - HackDevengers 1.0 Hackathon; built and pitched a working prototype under a fixed time constraint.
- Open Source:** 20+ public GitHub repositories spanning AI/ML, systems programming, and full-stack engineering.
- Certifications:** Google AI Essentials; Introduction to Artificial Intelligence (IBM); Git and GitHub (Google); Copilot Agent Mode (GitHub).